COVID-19 Data Analysis Project

1. Project Overview

This project analyzes global COVID-19 data using Python. It explores reported cases, deaths, testing, vaccination progress, and country-level differences through data analysis and visualization.

The project also focuses on Bangladesh to examine its COVID-19 case trends, death trends, and vaccination progress.

2. Project Objectives

- Analyze global COVID-19 cases and deaths over time.
- Identify countries with high reported case and death counts.
- Explore Bangladesh's COVID-19 situation.
- Analyze vaccination progress in Bangladesh and other countries.
- Calculate and compare Case Fatality Rates (CFR).
- Examine relationships between selected COVID-19 indicators.
- Present findings through statistical summaries and visualizations.

3. Dataset Description

The dataset was obtained from the Our World in Data COVID-19 compact dataset.

Dataset Information	Value
Total records after preprocessing	629,868

Dataset Information	Value
Total columns after preprocessing	60
Unique countries/entities	262
First available date	2020-01-01
Last available date	2026-10-18

The dataset includes information about reported COVID-19 cases, deaths, testing, hospitalizations, vaccinations, population, and selected demographic indicators.

4. Tools and Technologies

- Python
- NumPy
- Pandas
- Matplotlib
- Seaborn
- Jupyter Notebook

5. Data Preprocessing

The preprocessing workflow included:

- Loading the COVID-19 CSV dataset.
- Inspecting dataset dimensions, data types, and missing values.
- Removing completely empty columns.
- Checking missing country and date values.
- Inspecting duplicate country-date records.
- Sorting records chronologically.
- Reviewing negative daily case and death values.

- Creating derived date and statistical features.

Missing values were not universally replaced with zero because a missing observation does not necessarily mean zero reported cases.

6. Exploratory Data Analysis

The exploratory analysis examined:

- Dataset structure and descriptive statistics.
- Country-level cumulative cases and deaths.
- Global daily and monthly case trends.
- Global daily death trends.
- Bangladesh's daily and monthly case trends.
- Bangladesh's reported death trends.
- Bangladesh's vaccination progress.
- Relationships between selected numerical indicators.

7. Feature Engineering

The analysis included derived indicators such as:

- Case Fatality Rate (CFR).
- Year, month number, and month name.
- Cases excluding reported deaths.

Cases excluding reported deaths should not be interpreted as actual active cases or recovered cases.

8. Statistical Analysis

The statistical analysis included:

- Mean, median, standard deviation, minimum, and maximum.
- Country-level cumulative case and death comparisons.

- Case Fatality Rate calculations.
- Cases and deaths per million people.
- Full vaccination coverage comparisons.
- Pearson correlation analysis.
- Spearman rank correlation analysis.

9. Key Findings

The project findings are generated from the processed dataset and are documented in the accompanying CSV reports.

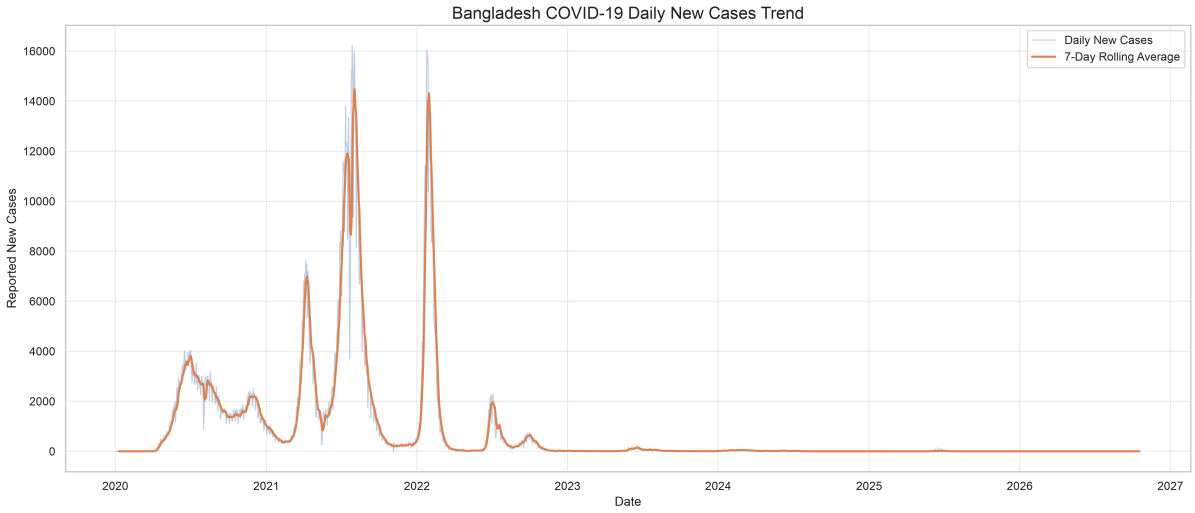
See the following files in the reports directory:

- global_covid_findings.csv
- top_10_cases_findings.csv
- top_10_deaths_findings.csv
- correlation_findings.csv
- bangladesh_vaccination_findings.csv
- cfr_findings.csv

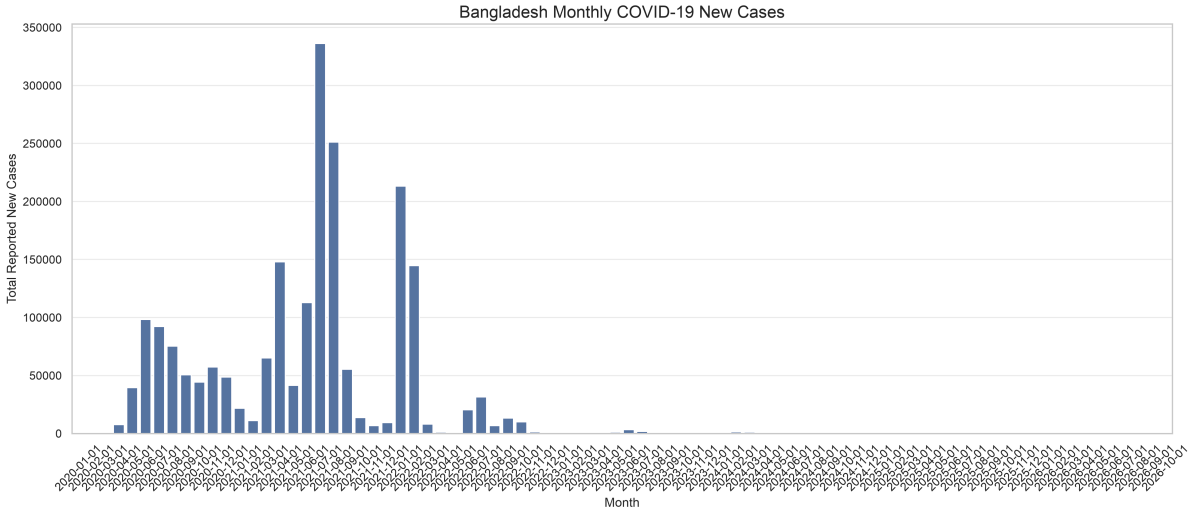
10. Visualizations

The following charts illustrate the major trends and comparisons identified during the COVID-19 data analysis.

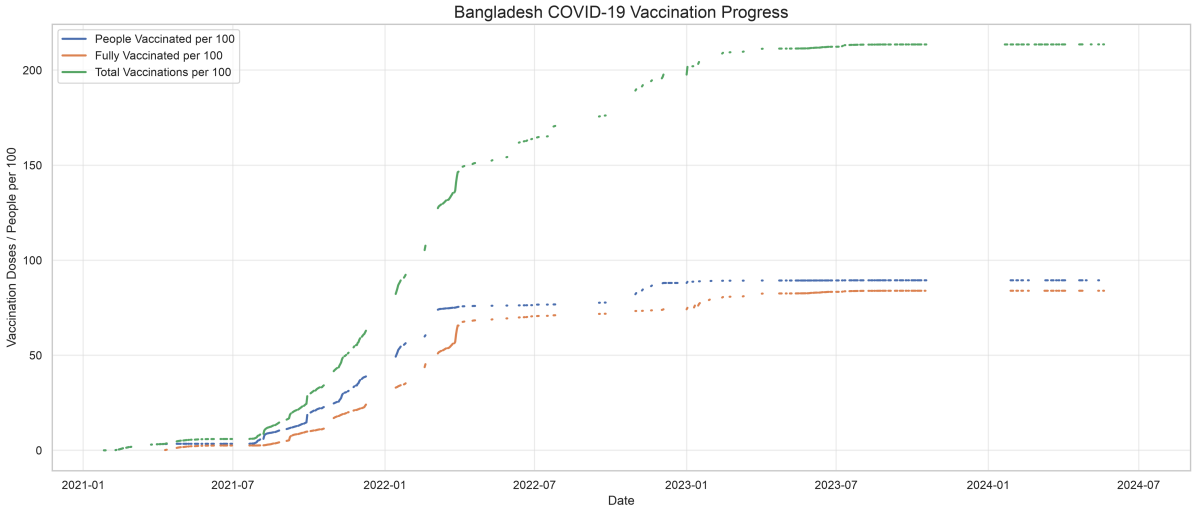
Bangladesh Cases Trend



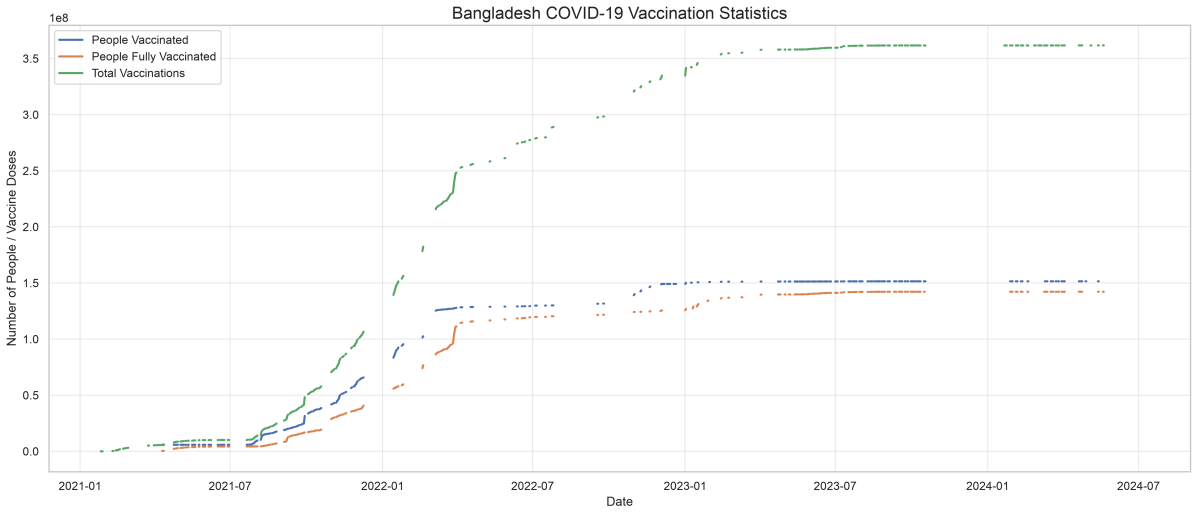
Bangladesh Monthly Cases



Bangladesh Vaccination Progress



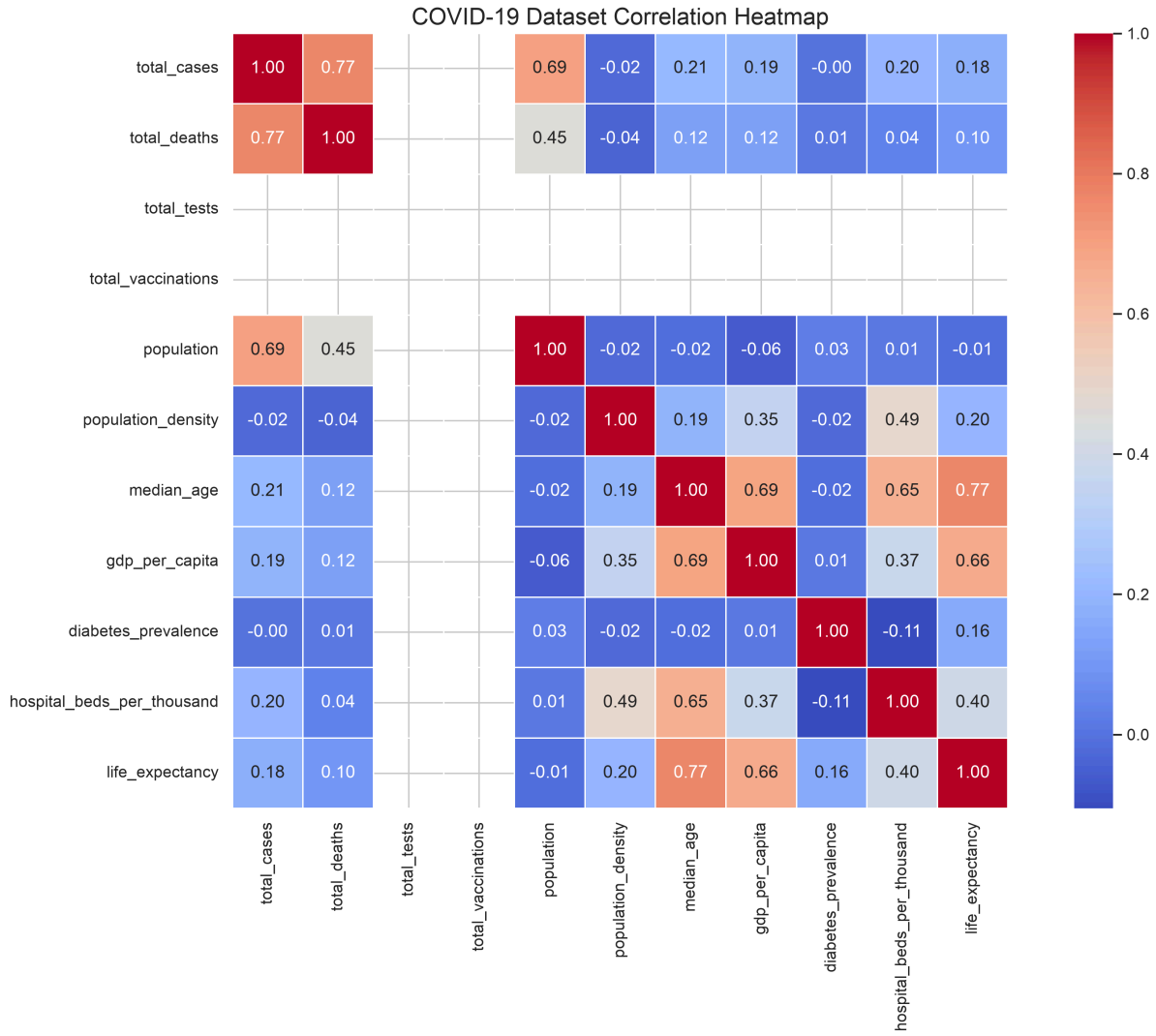
Bangladesh Vaccination Statistics



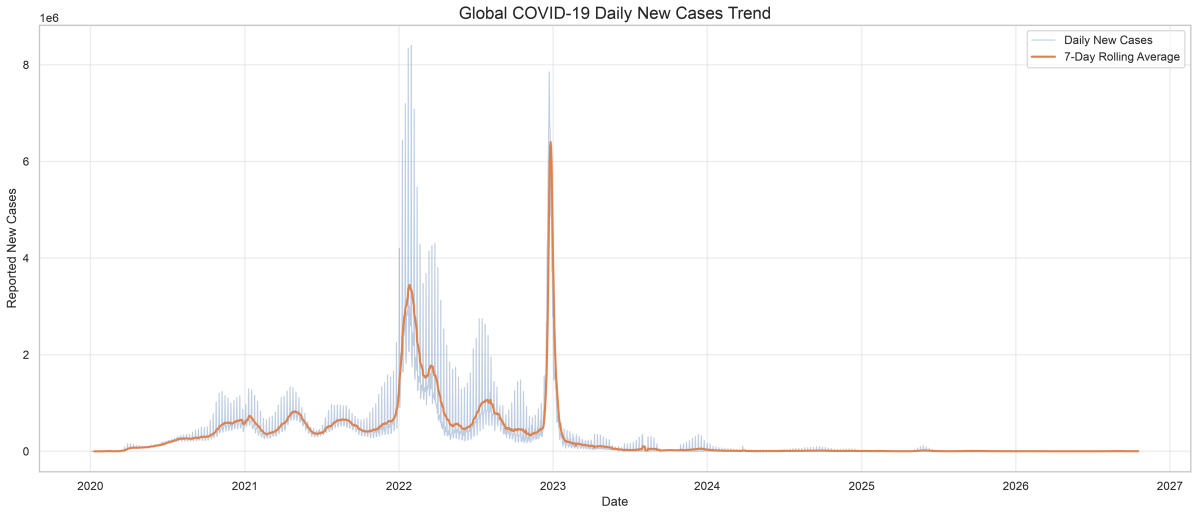
Cases Vs Deaths Relationship



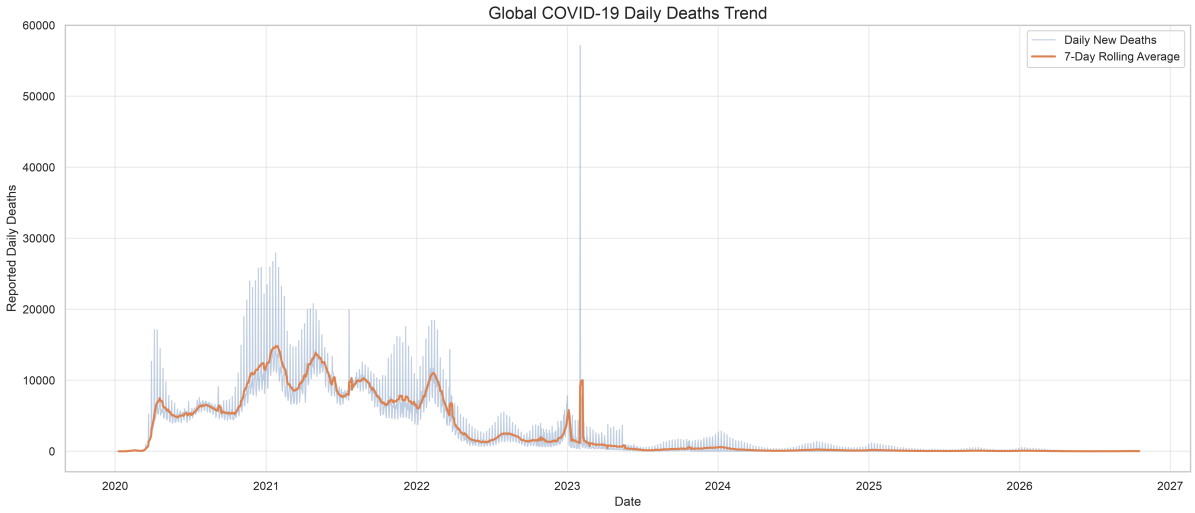
Covid Correlation Heatmap



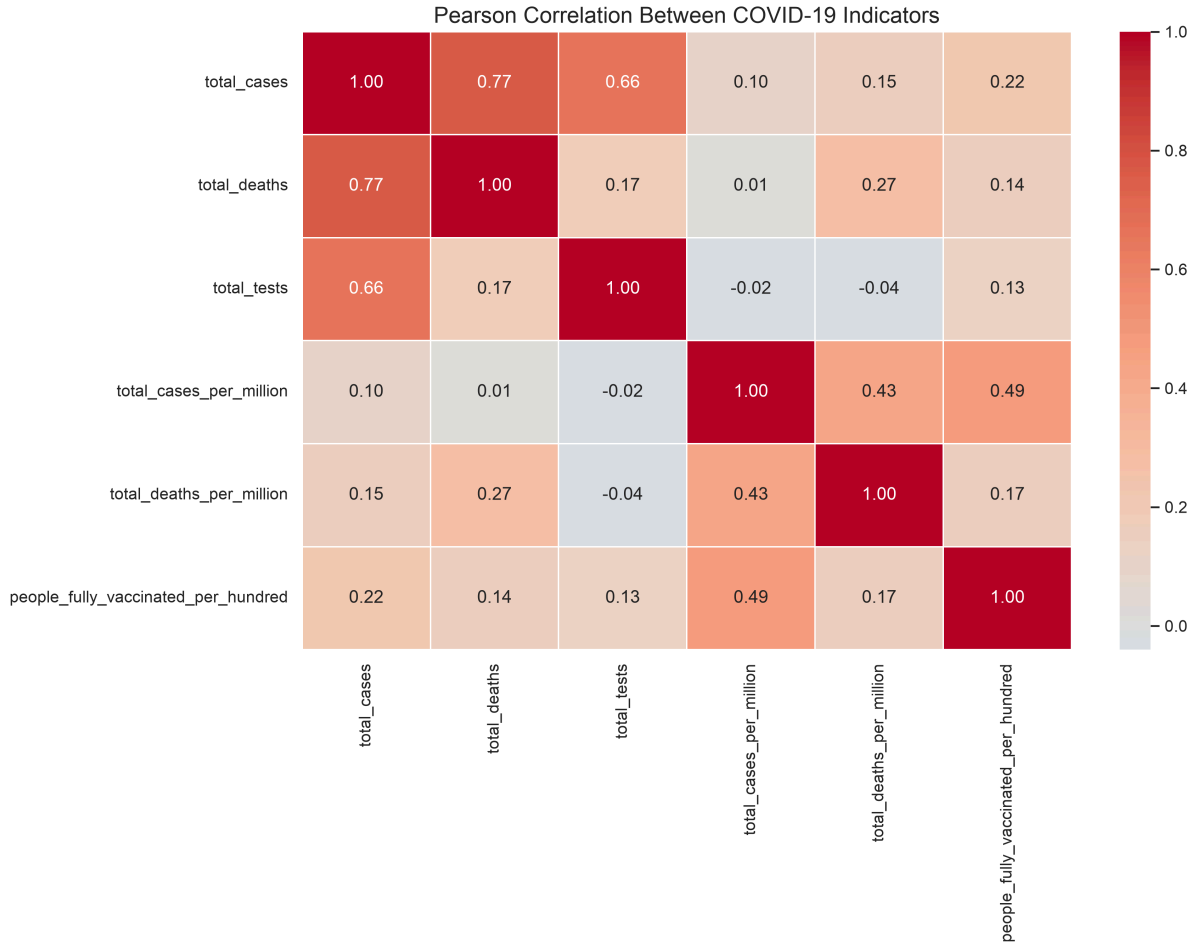
Global Cases Trend



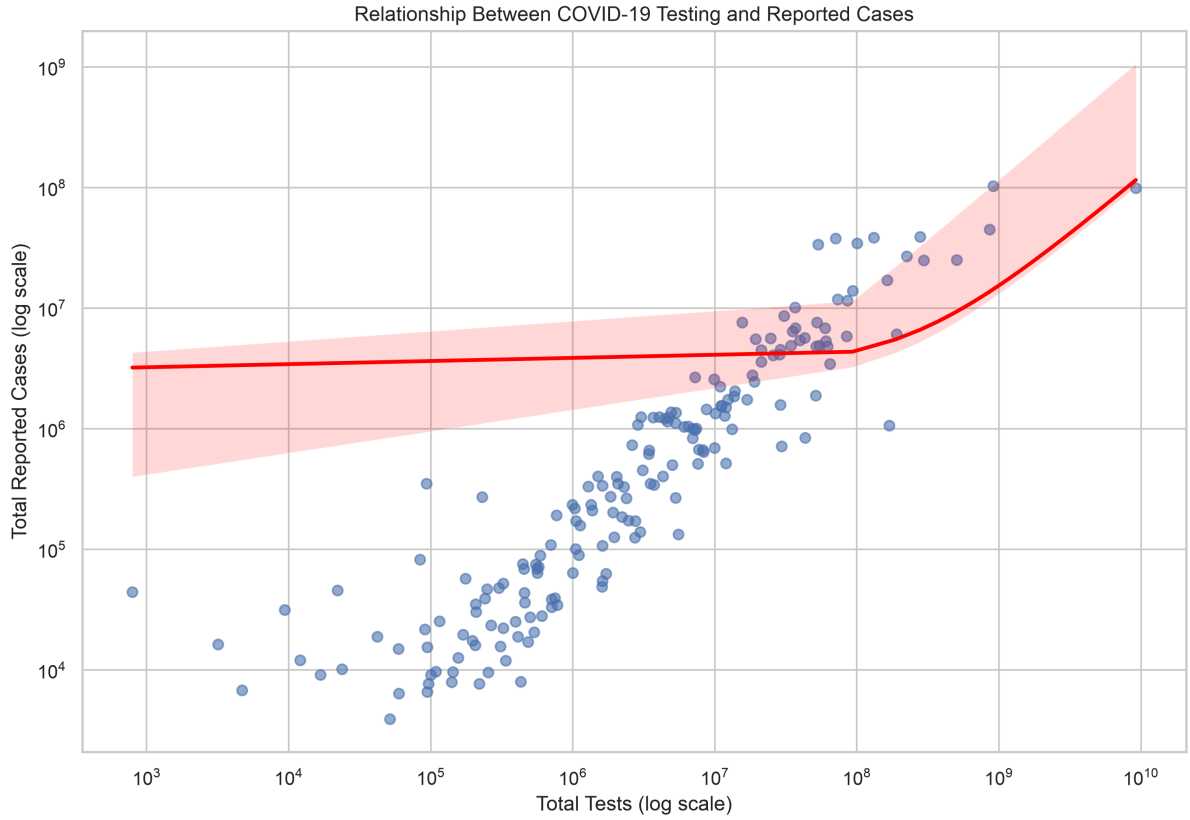
Global Deaths Trend



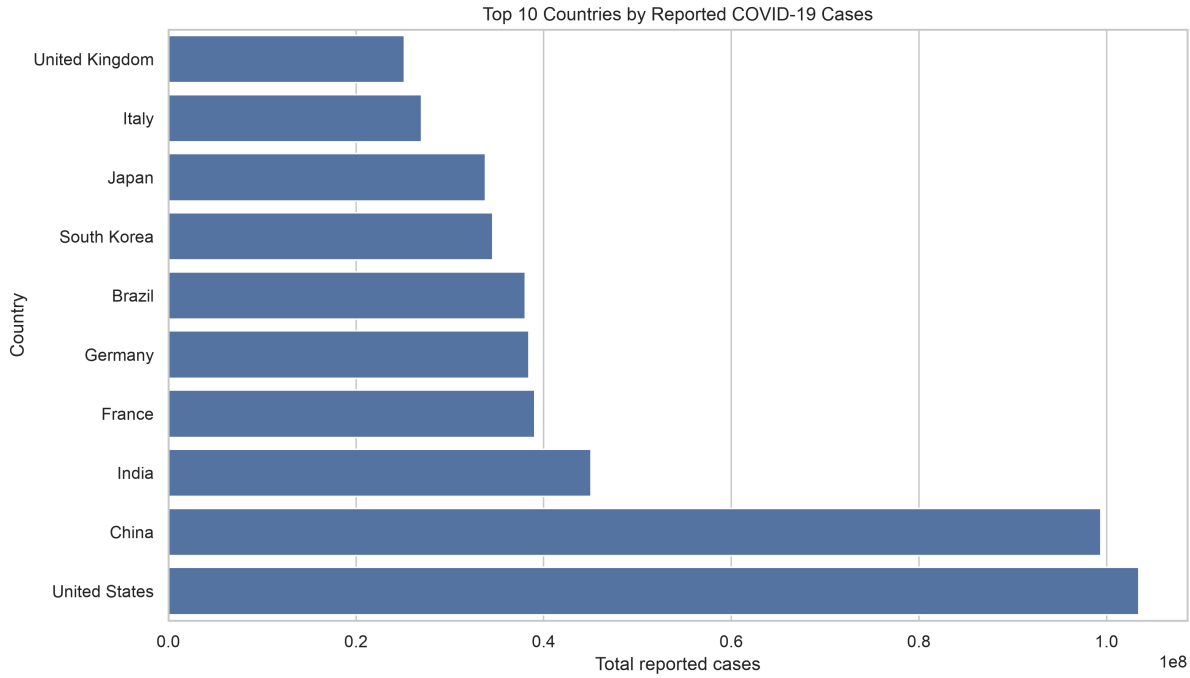
Pearson Correlation Heatmap



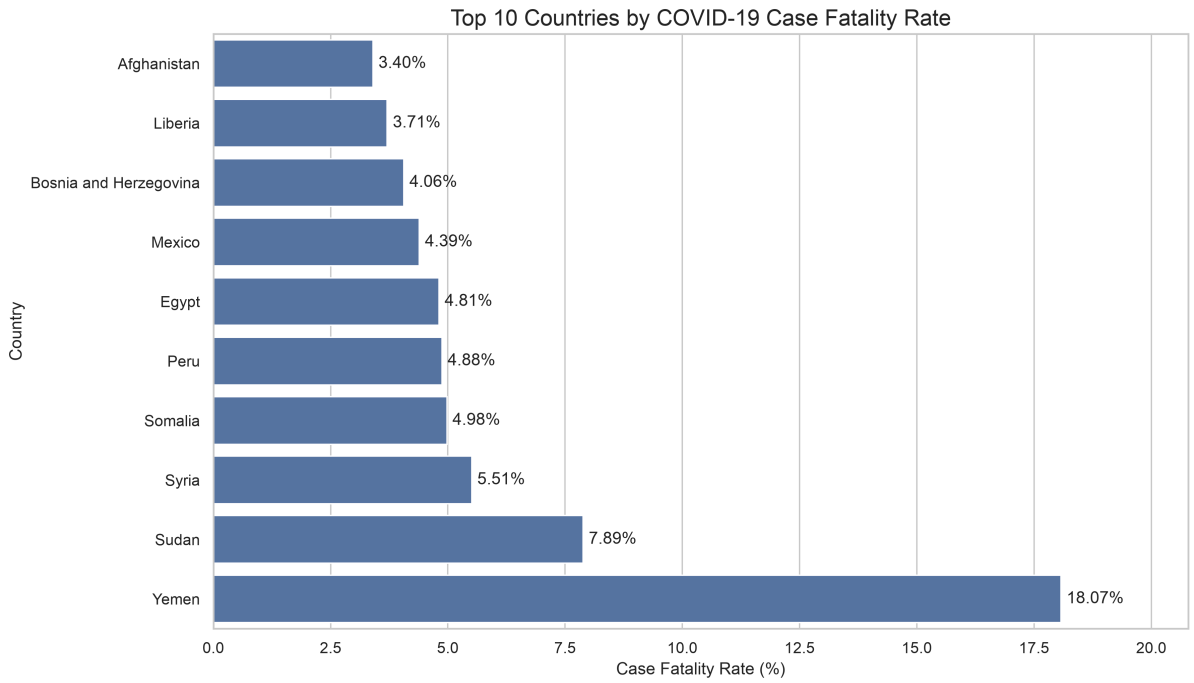
Testing Vs Cases Relationship



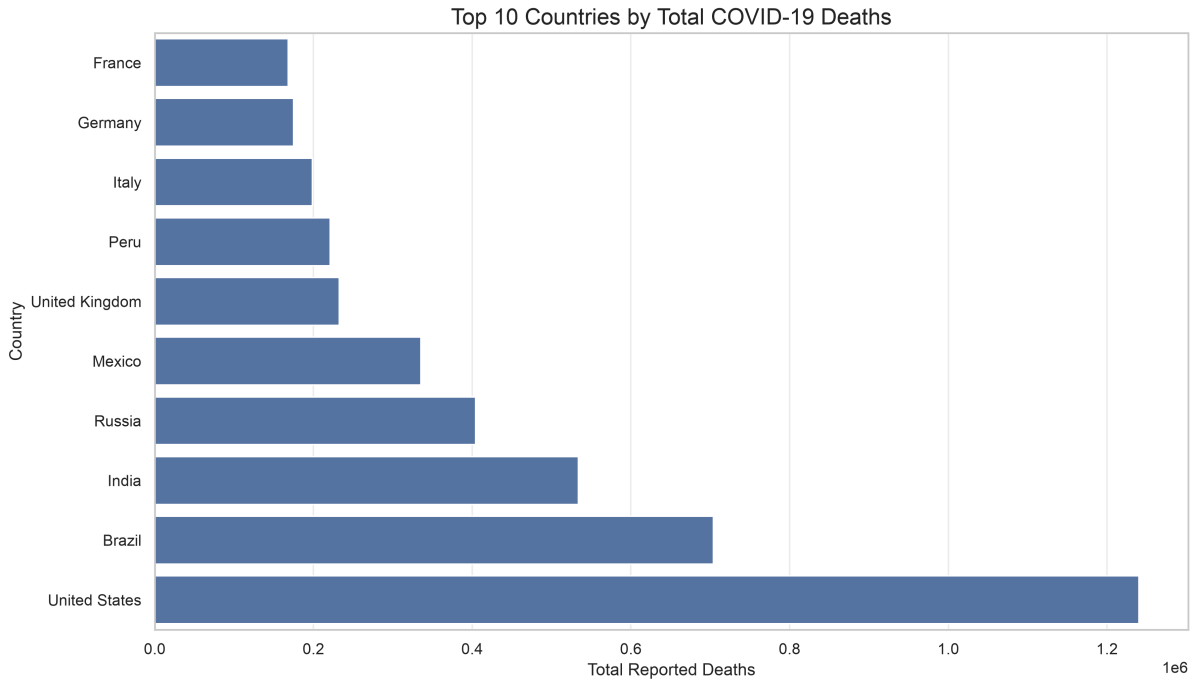
Top 10 Countries Cases



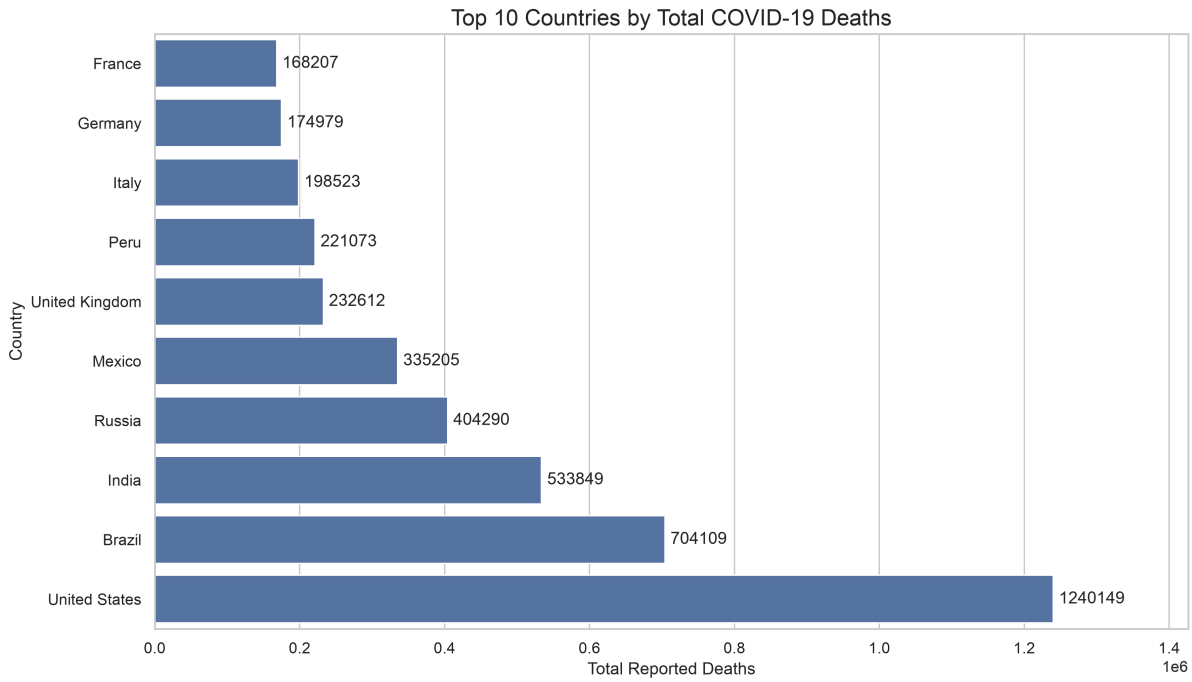
Top 10 Countries Cfr



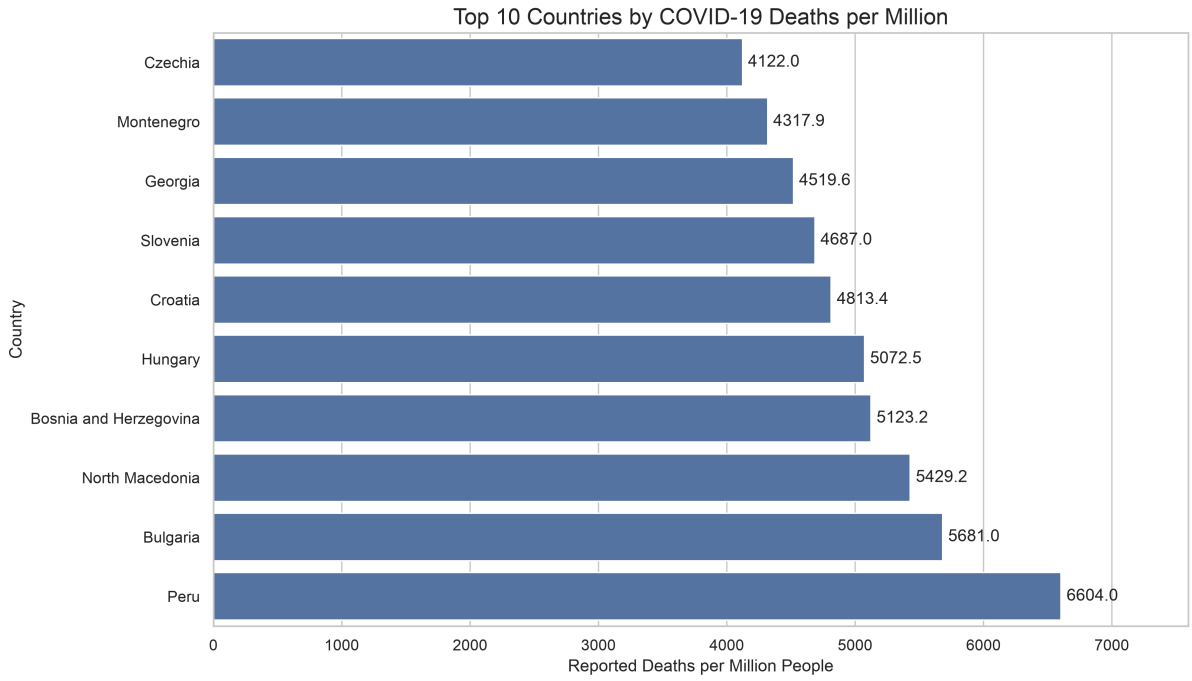
Top 10 Countries Deaths



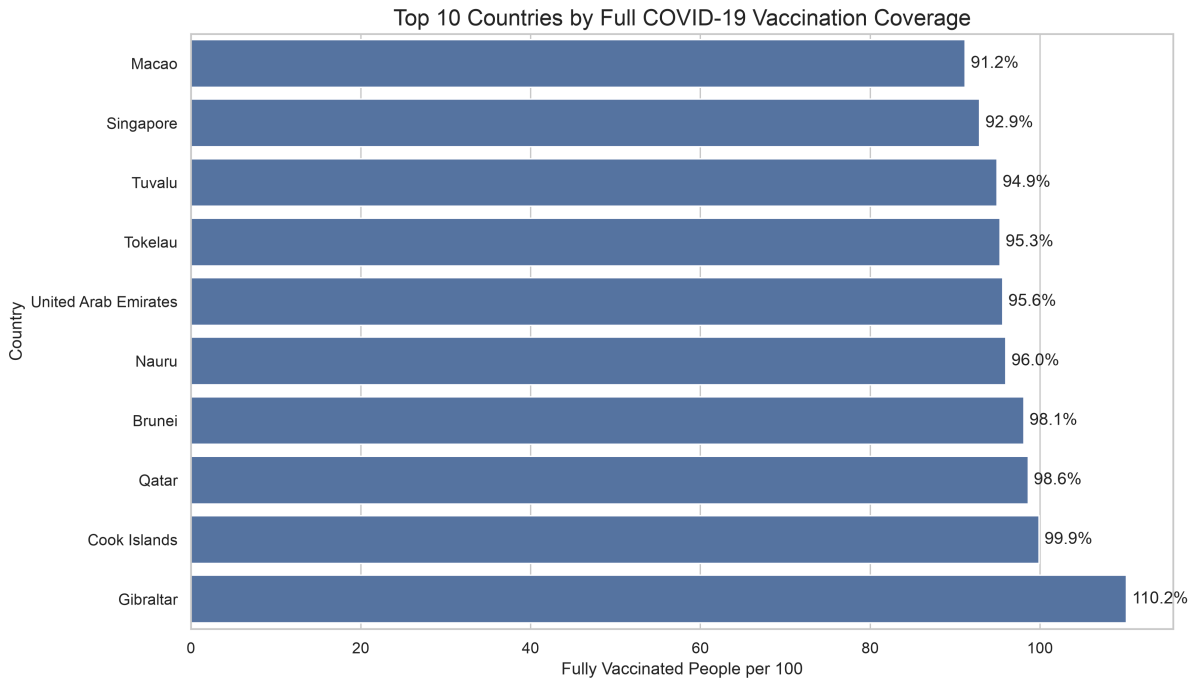
Top 10 Countries Deaths Labeled



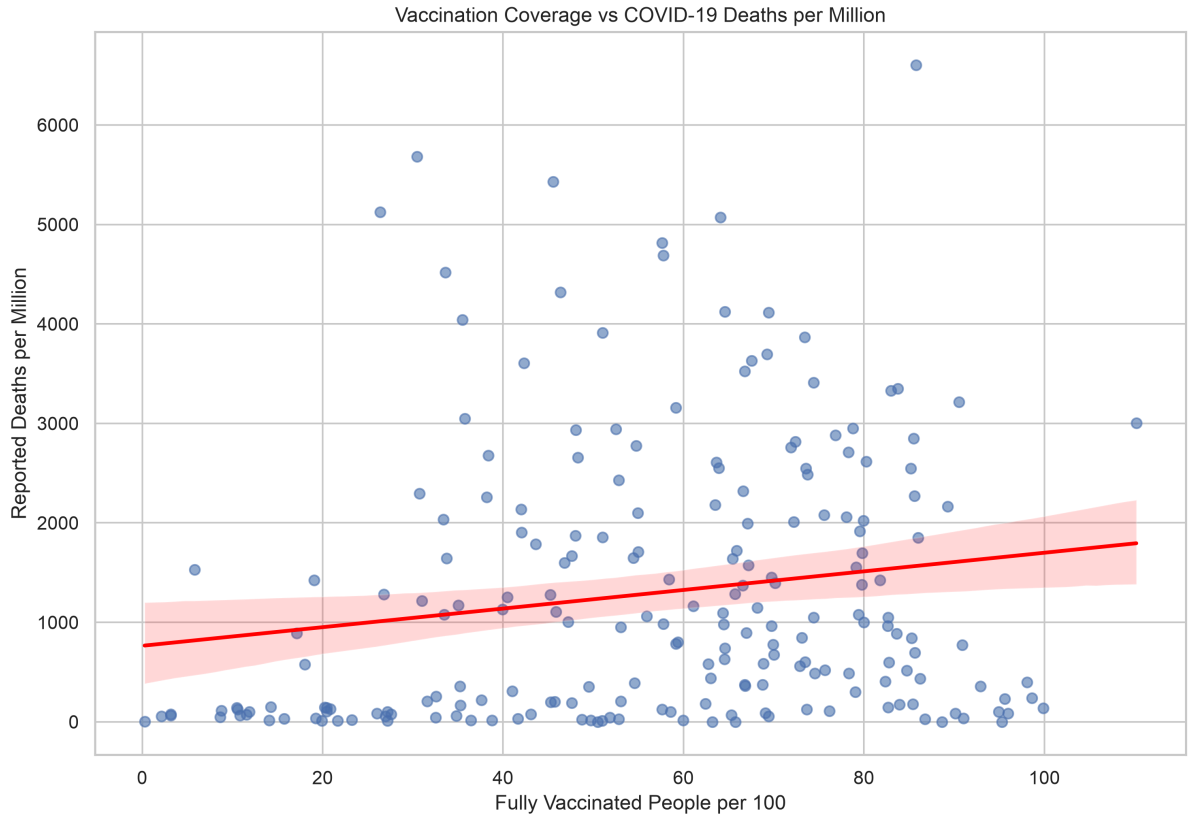
Top 10 Deaths Per Million



Top 10 Vaccination Coverage



Vaccination Vs Deaths Relationship



11. Limitations

- Reported cases and deaths depend on testing and reporting practices.

- Missing values can affect country-level comparisons.
- Countries may have different latest available observation dates.
- Cumulative totals should not be interpreted as daily counts.
- Negative daily values may reflect retrospective reporting corrections.
- Correlation does not establish causation.
- Country-level vaccination and death comparisons do not prove vaccine effectiveness.
- Per-million indicators help account for population size but do not remove differences in demographics, healthcare systems, or reporting quality.

12. Conclusion

This project demonstrates how Python data analysis tools can be used to process, explore, visualize, and statistically analyze a large real-world public health dataset.

The analysis provides an overview of global COVID-19 trends and a focused examination of Bangladesh. It also demonstrates the importance of data quality, appropriate statistical interpretation, and clear visual communication.

Project: COVID-19 Data Analysis
Dataset: Our World in Data COVID-19 Compact Dataset
Environment: Python and Jupyter Notebook